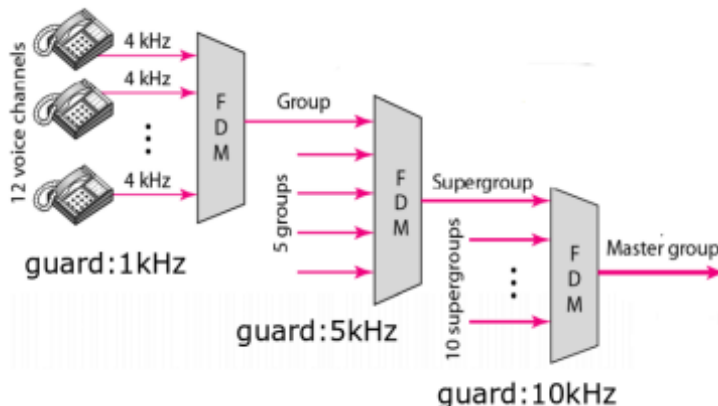


Chapter 6

Frequency Division Multiplexing (FDM):

1. Assume five 20KHz channels are multiplexed using FDM. If the bandwidth of each guard band is 5KHz, calculate the output channel Bandwidth
2. Assume we have an output channel bandwidth of 100 KHz starting from 1250KHz. There are 5 input channels. If the guard band between two channels is 5KHz, what will be the maximum bandwidth for each channel that we can assign?
3. Assume we have an output channel bandwidth of 100 KHz starting from 1250KHz. Assume five 20KHz channels are multiplexed using FDM. Calculate the bandwidth of the guard band.
4. The Following **FDM hierarchy** has been used by a telephone company. How many voice channels can be multiplexed together in the master group? What is the required bandwidth for the multiplexing?



Time Division Multiplexing (TDM):

1. Five channels are multiplexed using TDM. Each channel can process 1000 bits per second. if 5 bits at a time are to be multiplexed using the concept of TDM with 2 synchronizing bits, then answer the following questions.
 - a. Input rate & input duration
 - b. Frame rate and frame duration
 - c. Output rate and output duration
2. Five channels are multiplexed using TDM. Each channel can process 1000 pages per 5 seconds. Each page has 20 lines with 40 characters. if 5 lines at a time are to be multiplexed using the concept of TDM with 5 synchronizing bits, then answer the following questions.
 - a. Input rate & input duration
 - b. Frame rate and frame duration
 - c. Output rate and output duration

3. Five channels, three with a bit rate of 240 kbps and two with a bit rate of 220 kbps, are to be multiplexed with one synchronization bit. answer the following questions.
- Input rate & input duration
 - Frame rate and frame duration
 - Output rate and output duration
4. Six channels, three with a bit rate of 240 kbps and three with a bit rate of 120 kbps, are to be multiplexed. If 5 characters from each channel are to be multiplexed with three synchronization bits. answer the following questions.
- Input rate & input duration
 - Frame rate and frame duration
 - Output rate and output duration
 - How many input channels are there after doing multiplexing?
5. Five channels, two with a bit rate of 240 kbps and three with a bit rate of 180 kbps, are to be multiplexed with one synchronization bit. answer the following questions.
- Input rate & input duration
 - Frame rate and frame duration
 - Output rate and output duration
 - How many input channels are there after doing multiplexing?

Chapter 10

Block Coding:

1. If the 2 bit datawords are converted to the following codewords, for how many bits of error can we successfully detect and correct for this scheme?

Dataword	Codeword
00	11100000
01	00011101
10	10101010
11	01010111

2. Suppose two channels are using the following block coding scheme for error direction and correction. The receiver received the codeword 10101010 that contains error. Determine if the original dataword can be retrieved from it or not?

Dataword	Codeword
00	11110000
01	00001101
10	10111010
11	01110111

Cyclic Redundancy Check (CRC):

1. Suppose you want to transmit the message 1011101000 and protect it from errors using the CRC generator polynomial 11011. Using binary division, show the message that should be transmitted.

Later, corrupt the right-most sixth bit of the transmitted message and show that the error is detected by the receiver using CRC technique.

2. Suppose you want to transmit the message $x^8 + x^7 + x^5 + x^4 + x + 1$ and protect it from errors using the CRC generator polynomial $x^4 + 1$. Using **polynomial** division, show the message that should be transmitted.

Later, corrupt the right-most fifth bit of the transmitted message and show that the error is detected by the receiver using CRC technique.